Probability, Statistics and Optimization Techniques
Mini Project Report

on

Simulation of Web Series Genre Preferences Among Youth

Contents

1. Introduction

Project Title and Context

This project focuses on analyzing and simulating web series genre preferences among youth using Probability, Statistics, and Machine Learning techniques. With the rapid growth of OTT platforms such as Netflix, understanding viewer preferences has become crucial for content recommendation and strategic decision-making.

Objective of the Analysis

- To identify popular web series genres among youth
- To analyze viewing patterns using statistical methods
- To simulate user preference behavior
- To build a machine learning model to predict user engagement

Scope and Limitations

The study is based on a Netflix dataset and simulated user behavior. Since real user demographic data is not available, certain assumptions are made, which may affect real-world accuracy.

2. **Problem Statement**

The objective of this project is to analyze and predict web series genre preferences among youth using statistical and machine learning techniques.

This problem is important because OTT platforms rely heavily on user preferences to improve content recommendations, increase user engagement, and optimize viewing experience.

3. **Data Collection**

Sources of Data

The dataset was obtained from Kaggle (Netflix Content Dataset).

Data Description

The dataset includes:

- Type (Movie/TV Show)
- Genre (listed_in)
- Release Year
- Rating

Additional simulated features:

- Age (16–30)
- Watch Time
- Preference Score

Data Quality Issues

- Missing values in some columns
- No direct user preference data
- Bias due to simulated features

4. **Data Preprocessing**

Cleaning Steps

- Removed rows with missing values in important columns
- Selected only relevant features
- Extracted primary genre from multiple categories

Transformation

- Created new feature: Genre
- Simulated Age and Watch Time
- Generated Preference Score using probability distribution

Tools Used

Python (Pandas, NumPy)

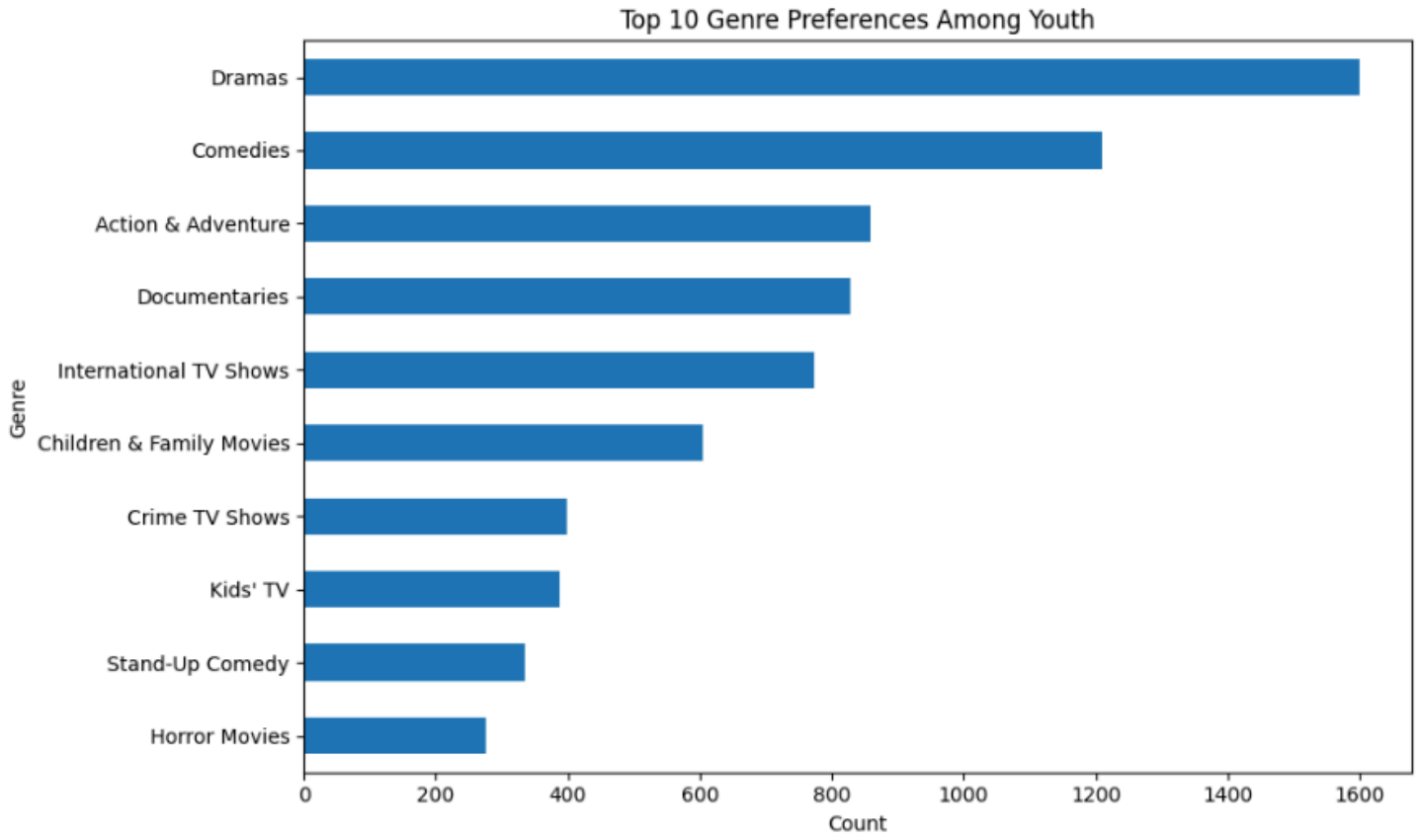
5. **Exploratory Data Analysis (EDA)**

Summary Statistics

- Mean Age ≈ 22 years
- Median Watch Time ≈ 5 hours
- Variance indicates moderate variation in viewing behavior

Visualizations

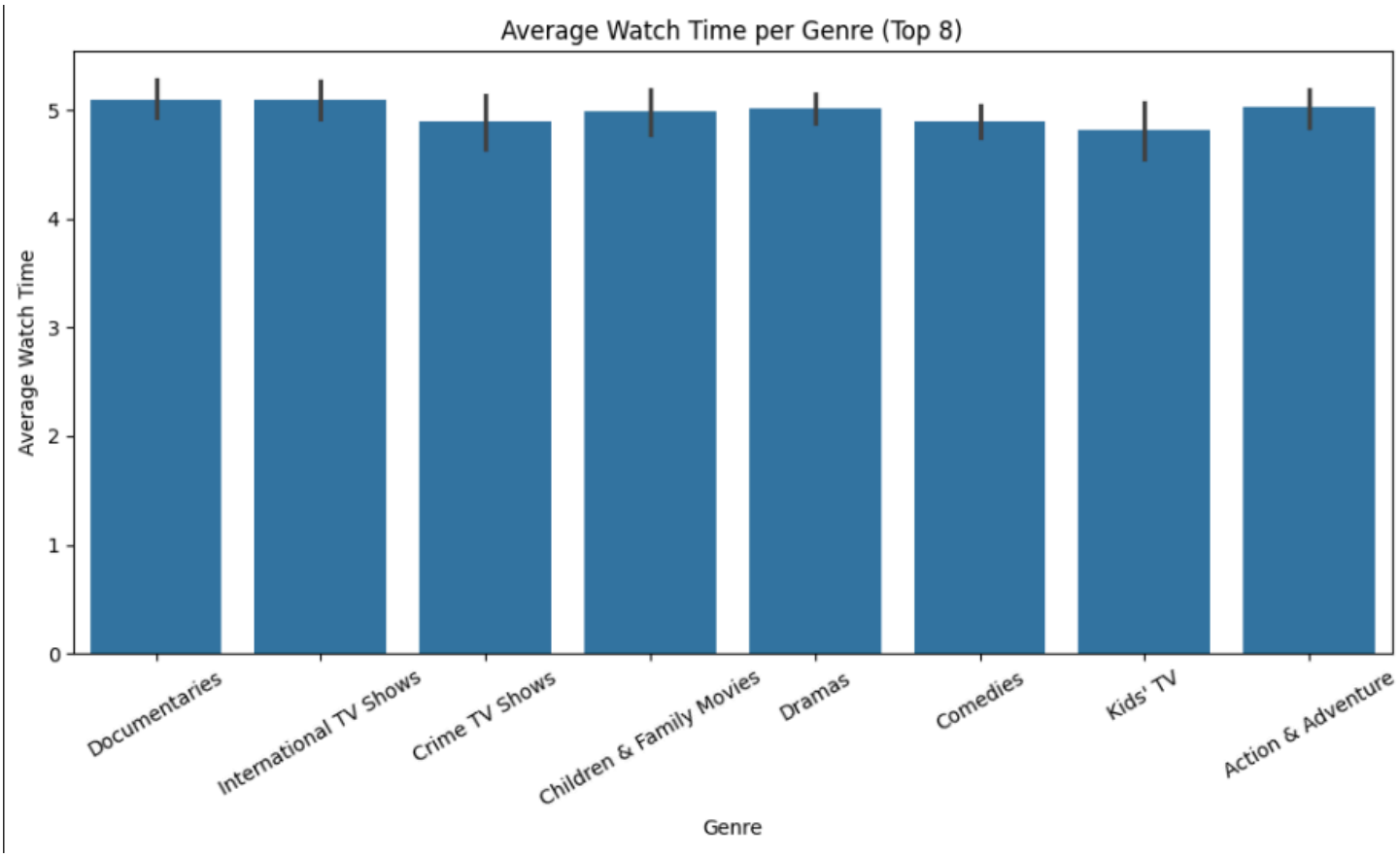
- Genre Distribution (Top 10)



This graph shows the most popular genres among youth. Drama and Comedy are the most preferred, indicating higher interest in these categories.

It also shows that a few genres dominate overall viewing preferences.

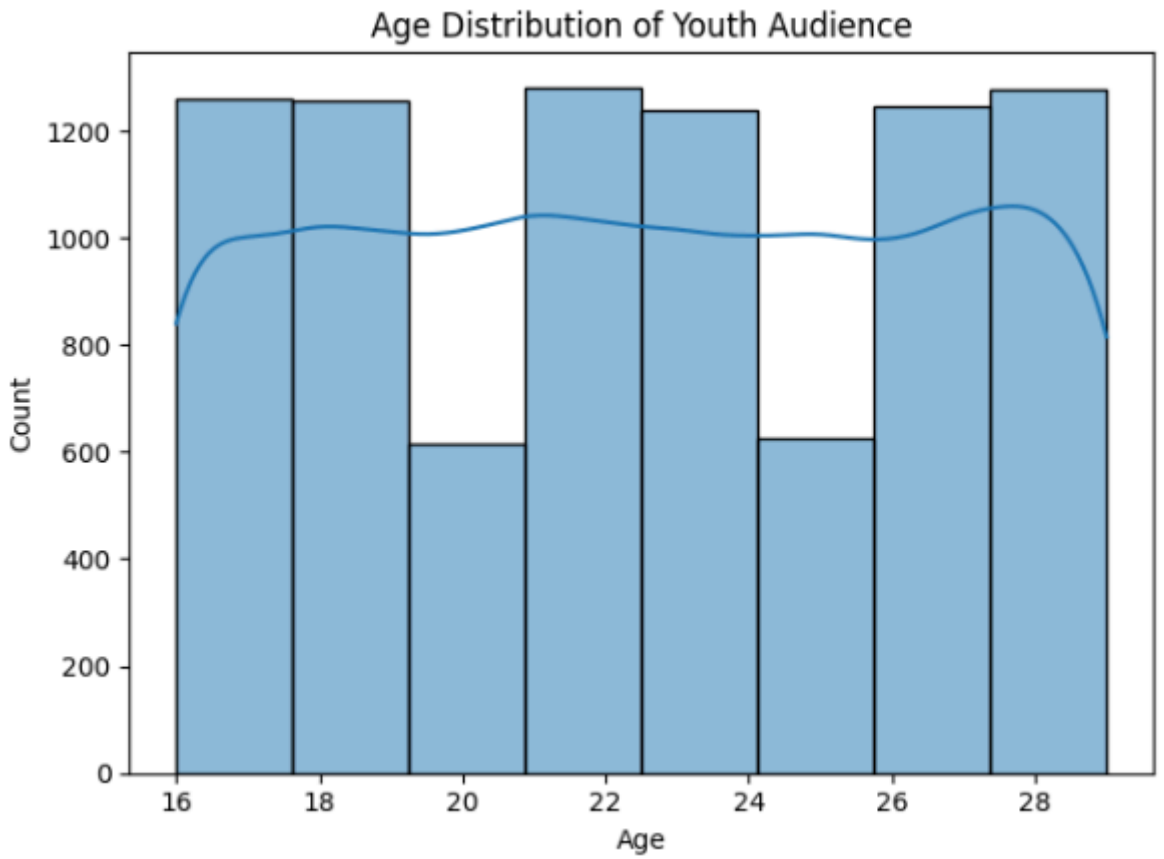
- Watch Time vs Genre



This graph shows the average watch time for different genres. Some genres have higher watch time, indicating stronger user engagement.

It suggests that certain types of content are more binge-worthy than others.

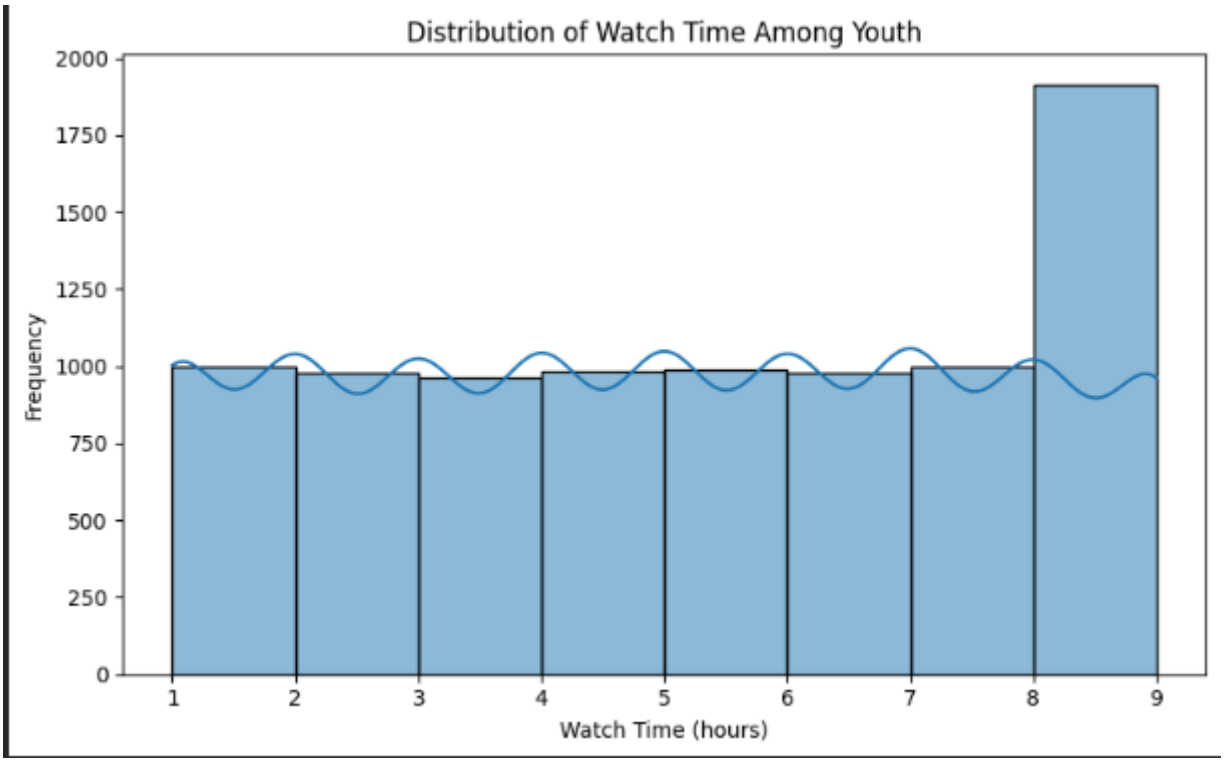
- Age Distribution



The graph shows that most users fall within the age group of 18–25 years.

This confirms that the dataset mainly represents youth audience and validates the scope of the project.

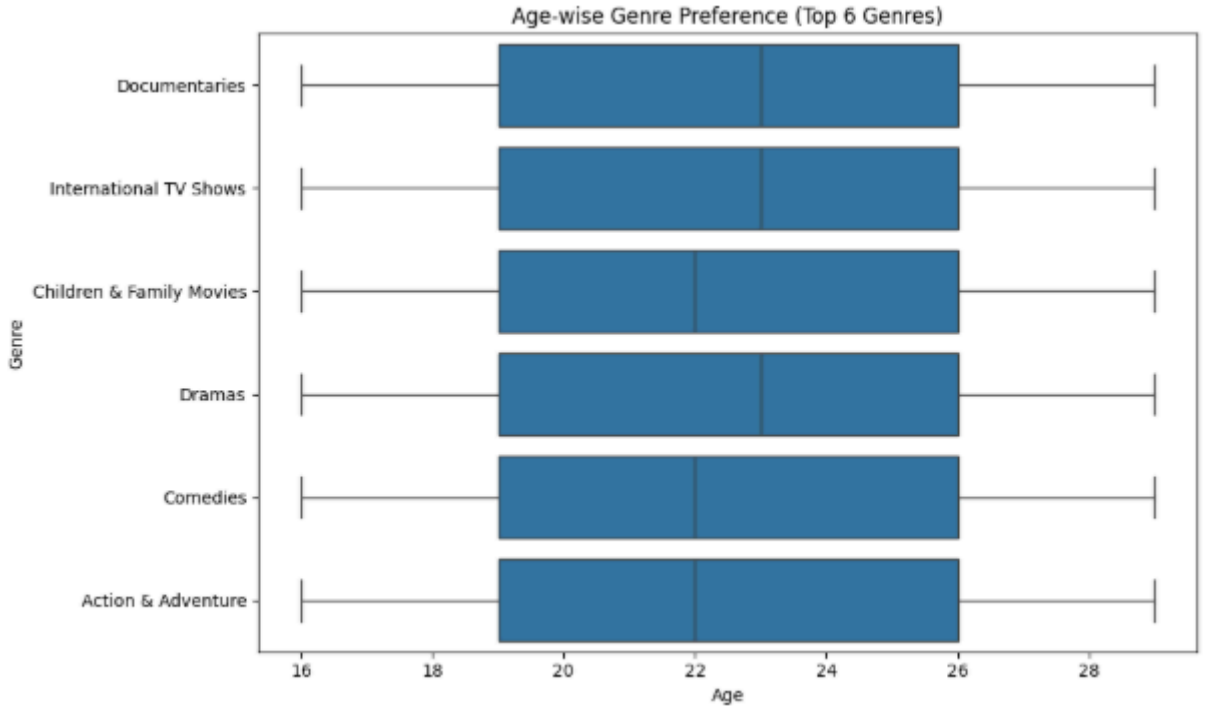
- Watch Time Distribution



This graph shows how watch time is spread among users. Most users have moderate watch time, while some show higher engagement.

It indicates variation in viewing habits among youth.

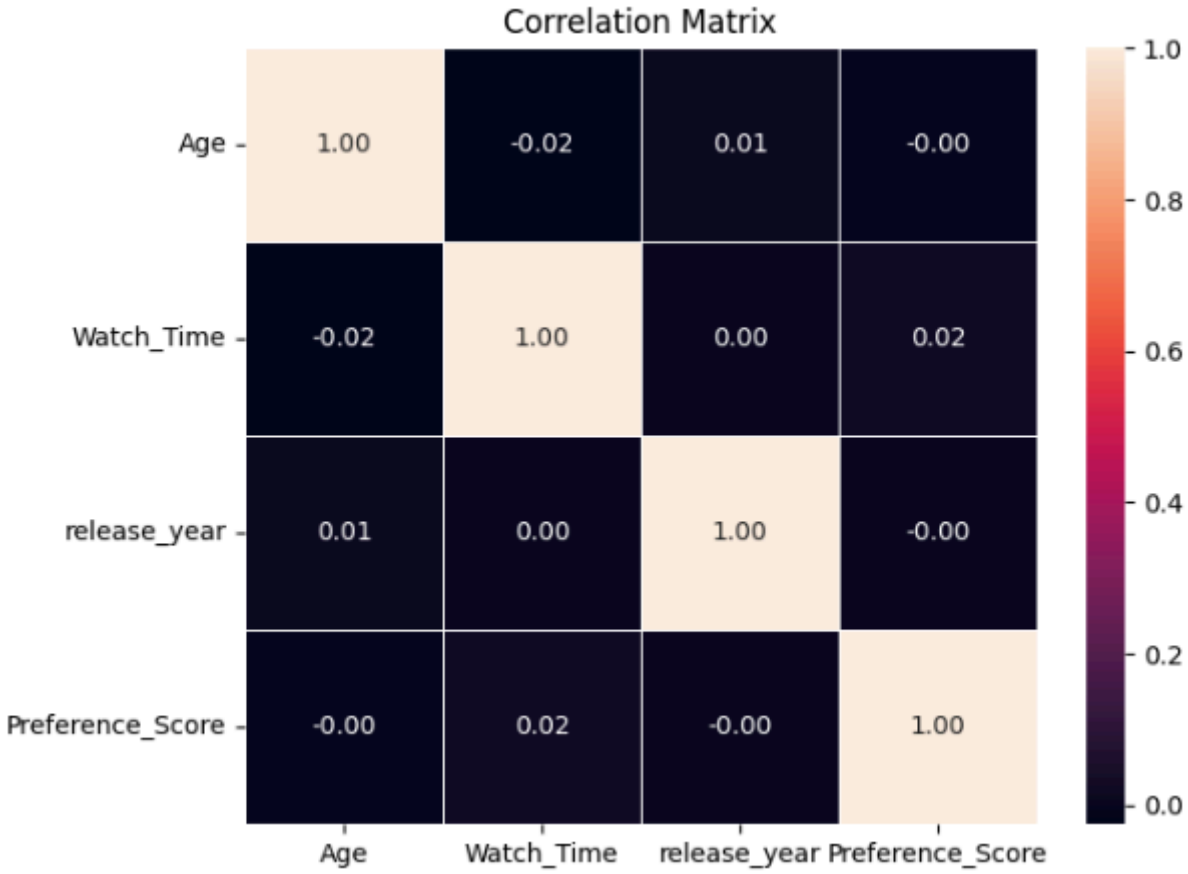
- Age vs Genre (Boxplot)



This boxplot shows how age varies across different genres. Most genres have users in a similar age range, but slight differences indicate variation in preferences.

It helps understand which age groups prefer certain genres.

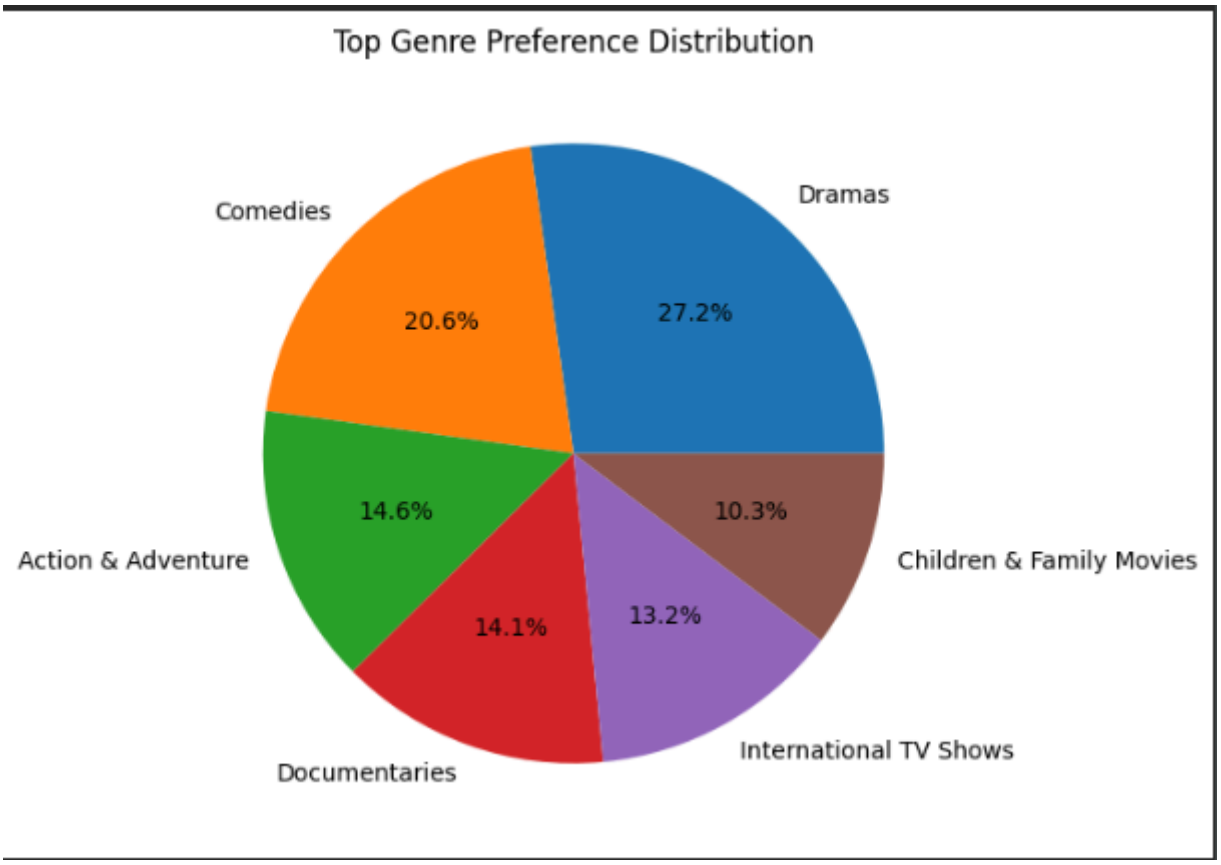
- Correlation Heatmap



The heatmap shows relationships between variables like age, watch time, and release year.

The correlations are weak, indicating that these variables are not strongly dependent on each other.

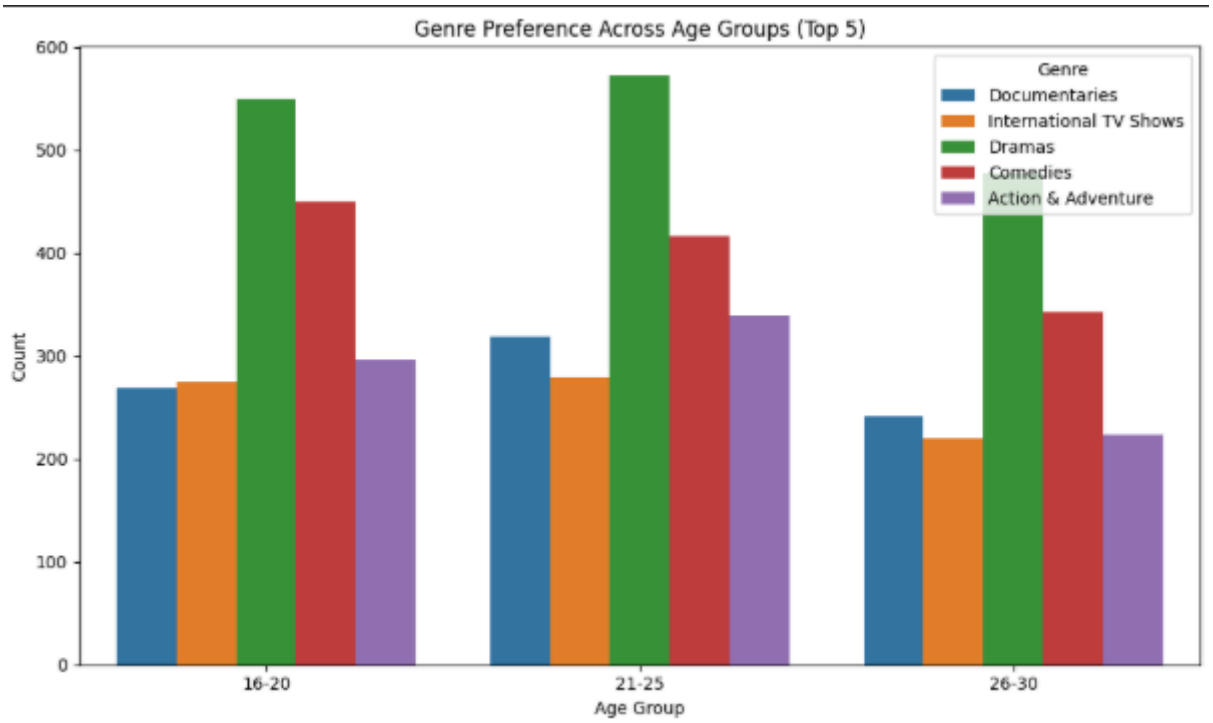
- Probability Pie Chart



This pie chart represents the probability distribution of top genres.

It shows that a few genres contribute more to overall preferences, highlighting dominant viewing trends.

- Age Group vs Genre



This graph shows how genre preferences vary across different age groups.
It indicates that certain genres are more popular in specific age ranges within youth.

Key Patterns

- Drama and Comedy are most preferred genres
- Youth aged 18–25 show higher engagement
- Certain genres have higher watch time → more engaging

6. **Methodology**

Analytical Techniques

- Probability Distribution (Genre preference)
- Statistical Analysis (mean, variance, correlation)
- Machine Learning (Random Forest Classification)
- Simulation Techniques

Justification

These methods help in understanding patterns, predicting preferences, and modeling user behavior effectively.

Tools Used

- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn

7. **Results & Insights**

The analysis revealed that genres such as Drama and Comedy are the most preferred among youth. The probability distribution confirmed that a few genres dominate viewer preferences.

Age-wise analysis showed that the majority of viewers fall within the 18–25 range, indicating higher engagement in this group. Watch time analysis suggested that certain genres result in higher user engagement, making them more binge-worthy.

The machine learning model was used to classify user preferences into high and low categories. The model achieved an accuracy of approximately 49%, indicating moderate predictive performance due to the simulated nature of the dataset.

Feature importance analysis showed that release year and age have significant influence, while watch time contributes moderately to preference prediction.

8. **Discussion**

The results align with general OTT platform trends, where popular genres tend to dominate youth preferences. However, the use of simulated data introduces certain assumptions that may affect real-world accuracy.

The relatively moderate performance of the machine learning model highlights the complexity of user behavior and the need for richer datasets. Despite this, the model successfully identifies key influencing factors, making it useful for understanding viewing patterns.

The study demonstrates how statistical and machine learning techniques can be applied to analyze user preferences effectively.

9. **Conclusion**

This project successfully analyzed and simulated web series genre preferences among youth using Probability, Statistics, and Machine Learning techniques.

The study provided insights into genre popularity, user engagement, and influencing factors such as age and watch time. The integration of simulation techniques helped bridge the gap between available data and real-world user behavior.

The application of machine learning further enhanced the analysis by enabling prediction of user preferences. Overall, the project demonstrates the practical application of PSOT concepts in real-world scenarios such as content recommendation systems

10. **Appendices**

a) Data Loading

```
df = pd.read_csv("netflix_titles.csv")
```

b) Data Preprocessing

```
df = df.dropna(subset=['listed_in', 'rating'])

df = df[['type', 'listed_in', 'release_year', 'rating']]

df['Genre'] = df['listed_in'].apply(lambda x: x.split(',')[0])
```

c) Simulation of Youth Data

```
df['Age'] = np.random.randint(16, 30, size=len(df))

df['Watch_Time'] = np.random.randint(1, 10, size=len(df))
```

d) Preference Score Generation

```
genre_weights = {

  'Drama': 0.3,

  'Comedy': 0.25,

  'Action': 0.2,

  'Thriller': 0.15,

  'Romance': 0.1

}
```

```
df['Preference_Score'] = df['Genre'].map(genre_weights).fillna(0.05)

df['Preference_Score'] += np.random.rand(len(df)) * 0.1
```

e) Machine Learning Model

```
df['Preference_Label'] = (df['Preference_Score'] > df['Preference_Score'].mean()).astype(int)


X = df[['Age', 'Watch_Time', 'release_year']]

y = df['Preference_Label']


X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)


model = RandomForestClassifier()

model.fit(X_train, y_train)


y_pred = model.predict(X_test)
```

10.2 Statistical Measures

The following statistical parameters were calculated:

- Mean Age
- Median Watch Time
- Variance of Watch Time
- Correlation between variables

STATISTICS

```
print("Mean Age:", df['Age'].mean())
print("Median Watch Time:", df['Watch_Time'].median())
print("Variance:", df['Watch_Time'].var())
```

```
Mean Age: 22.498693627172553
Median Watch Time: 5.0
Variance: 6.64681957719132
```

These measures helped in understanding central tendency, variability, and relationships within the dataset.

10.3 Machine Learning Evaluation Metrics

The following evaluation techniques were used:

- Accuracy Score

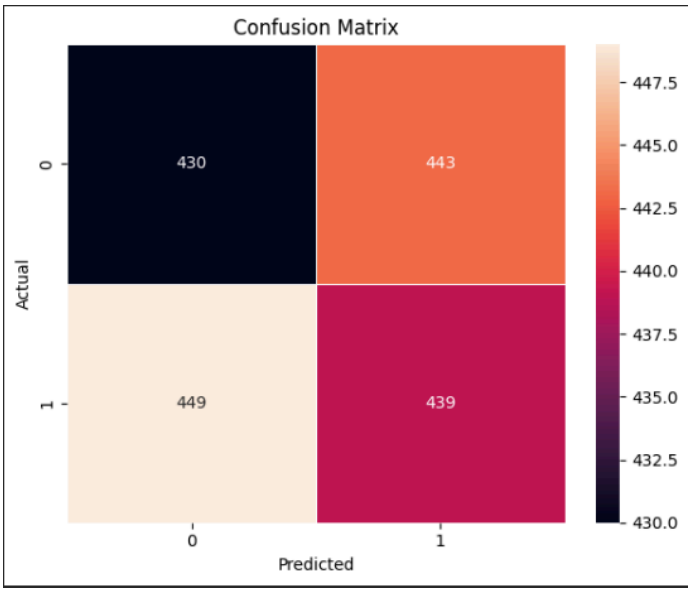
Accuracy

```
from sklearn.metrics import accuracy_score

accuracy = accuracy_score(y_test, y_pred)
print("Accuracy:", accuracy)
```

```
Accuracy: 0.4934696195343555
```

- Confusion Matrix: The confusion matrix shows the performance of the classification model. It compares actual and predicted values, indicating how well the model classifies user preferences.



These metrics were used to evaluate the performance of the classification model.

10.4 Feature Importance Analysis

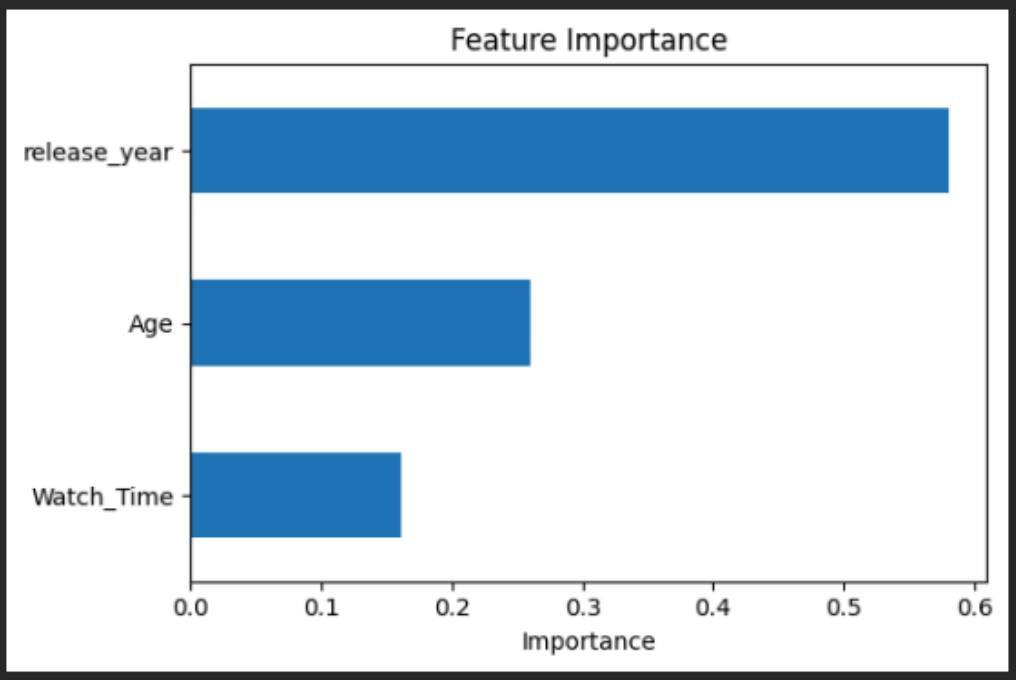
Feature Importance

```
import pandas as pd

importance = model.feature_importances_
features = X.columns

plt.figure(figsize=(6,4))
pd.Series(importance, index=features).sort_values().plot(kind='barh')

plt.title("Feature Importance")
plt.xlabel("Importance")
plt.tight_layout()
plt.show()
```



The feature importance graph represents the contribution of each input variable in predicting user preference. It was observed that *release year* has the highest importance, followed by *age*, while *watch time* has comparatively lower influence.

This indicates that newer content tends to attract more user attention, suggesting that recency plays a significant role in user preference. Age also contributes moderately, reflecting variation in preferences across different youth segments.

The relatively lower importance of watch time suggests that while it reflects engagement, it may not be a strong standalone predictor in this simulated dataset.

10.5 References

- Kaggle Dataset: Netflix Content Dataset
- Python Documentation (Pandas, NumPy, Matplotlib, Seaborn)
- Scikit-learn Documentation